Course Project – Part III

NUMN33/FMNN45(F): Numerical Methods for PDEs

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This is Part III of the overall course project which is based on DUNE and DUNE-FEM in your own Python environment.
In this part, we consider elliptic partial differential equations and investigate various properties. The project consists of both theoretical and practical exercises (tasks).

Please describe you solutions and observation in report form. For the practical exercises it should include

- a description of the problem considered,
- a description of your test cases in such a way so that they are reproducible, and
- a description of your tests, i.e. what is tested and why ...,
- your own interpretation of the results.
- Include important parts of your code as code snippets or append the entire code at the of the report.

Displaying code in Latex documents can, for example, be done like in this template: `https://gitlab.maths.lu.se/robertk/thesislatextemplate`. Many other examples can be found online.

Grading:

- All "non-living" and "non-generative AI" help is allowed, as long as proper references are given and you are able to explain the results during the oral exam. The use of generative AI is discouraged.
- **NUMN33:**
  - Minimum of 4 points in each section (theoretical/practical) and minimum 12 points in total for grade **G**.
  - Grade **G** and a minimum of 18 points total for a grade **VG**.
- **FMNN45:**
  - 4 points in each section (theoretical/practical) and minimum 12 points in total for a grade **3**.

- Grade **G** and a minimum 16 points total for a grade **4**.
- Grade **G** and a minimum 20 points total for a grade **5**.

- **FMNN45F:**
  - 6 points in each section (theoretical/practical) and minimum 16 points in total for grade **G**.

This assignment has **9 tasks** and **24 points** in total.

Theoretical Exercises

Task 1 (6 pts)

Consider the parabolic equation

$$\begin{aligned} &\tfrac{\partial}{\partial t}u = \Delta u - u && \text{in } (0,1)\times[0,T] \\ &\tfrac{\partial}{\partial x}u(0,t) = \tfrac{\partial}{\partial x}u(1,t) = 0 && \text{and } u(\cdot,0) = \eta \end{aligned} \tag{1}$$

1. Formulate the implicit Euler/FEM discretization of equation (1) with an equidistant grid in space, i.e., $h_i = h$ for all i. (2 pts)
2. Derive the linear equation system that needs to be solved in every time step. Make sure that you explicitly state the entries of the mass and stiffness matrices. (2+2 pts)

Task 2 (6 pts)

Prove that the Crank-Nicolson/FEM scheme converges as $\mathcal{O}(h^2+(t/n)^2)$ under the assumptions stated in Lecture 12 (10), i.e., prove the theorem on page 6, Lecture 12 (10).

Extra Task 3 (4 bonus pts, mandatory for FMNN45F!)

Consider the following abstract ODE in $\ell^2(\mathbb{R})$:(*)

$$\frac{\mathrm{d}}{\mathrm{d}t}u = Au, \quad u(0) = \eta, \tag{2}$$

where the linear operator $A : \mathcal{D}(A) \subset \ell^2(\mathbb{R}) \to \ell^2(\mathbb{R})$ is given by

$$\begin{aligned} \mathcal{D}(A) &= \{u \in \ell^2(\mathbb{R}) : Au \in \ell^2(\mathbb{R})\} \qquad \text{and} \\ Au &= \Big((Au)_1, (Au)_2, \ldots, (Au)_{2n-1}, (Au)_{2n}, \ldots\Big) \\ &= (u_2, -u_1, \ldots, nu_{2n}, -nu_{2n-1}, \ldots). \end{aligned}$$

*The space $\ell^2(\mathbb{R})$ consists of all real valued sequences $u = (u_1, u_2, \ldots)$ which are square summable, i.e., $\sum_{i=1}^{\infty} u_i^2 < \infty$, and $\ell^2(\mathbb{R})$ becomes a Hilbert space when equipped with the inner product $(u,v) = \sum_{i=1}^{\infty} u_i v_i$.

a) Prove that A is an unbounded operator.

b) Prove that equation (2) has a unique solution in $C^1\Big([0,T], \ell^2(\mathbb{R})\Big)$ when η is an element in $\mathcal{D}(A)$.

Practical Exercises

A nonlinear elliptic problem

$$-\nabla \cdot (K(u)\nabla u) = f \tag{3}$$

$f = 1$ is a forcing term and the diffusion tensor is

$$K(u) = (\varepsilon^2 + |\nabla u|^2)^{\frac{p-2}{2}}, \tag{4}$$

with ε being a small positive number. On the boundary we prescribe Dirichlet conditions $g = 0$.
We solve this problem in variational form

$$\int_\Omega K(u)\nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx.$$

on a domain $\Omega = [0,1]^2$.

Task 4 (2 pts)

- Setup grid for domain $\Omega = [0,1]^2$ and space a linear Lagrange space V_h.
- Define a discrete function u_h and also define constants for ε, p, f, and g. These can used as floats in UFL expressions but can be changed at a later stage without requiring any recompilation.
- Define a function $K(u)$ and set $\varepsilon = 10^{-5}$.
  Note: In UFL we write $|\nabla u|^2$ as `inner(grad(u),grad(u))`.
- Define the weak form given above.
- And finally, create a `galerkin` scheme and solve this system and store the result in u_h and also plot the result.

A parabolic problem

$$\partial_t u - \nabla \cdot K(\nabla u)\nabla u = f \tag{5}$$

$f = f(x,t)$ is a time dependent forcing term and the diffusion tensor is

$$K(u) = \frac{2}{1+\sqrt{1+4|\nabla u|}} \tag{6}$$

On the boundary we prescribe Neumann boundary conditions and require initial conditions $u(\cdot,0) = u_0(\cdot)$.
We solve this problem in variational form using Implicit Euler in time

$$\int_\Omega \frac{u^{n+1}-u^n}{\Delta t} v + K(u^{n+1})\nabla u^{n+1} \cdot \nabla v \, dx = \int_\Omega f(x,t^n+\Delta t) v \, dx + \int_{\partial\Omega} g(x,t^n+\Delta t) \cdot n \, v \, ds. \tag{7}$$

on a domain $\Omega = [0,1]^2$. We choose f, g so that the exact solution is

$$u_{ex}(x,t) = e^{-2t}\left(\frac{1}{2}(x^2+y^2) - \frac{1}{3}(x^3-y^3)\right) + 1.$$

Task 5 (2 pts)

Define constants

```
from dune.ufl import Constant
dt = Constant(0, name="dt")     # time step
t  = Constant(0, name="t")      # current time
```

After that implement the weak form `a` and right hand side `b` given in (7) using terms K, f, and g.
Note: Both f and g can be implemented in terms of u_{ex} and UFL, e.g.

```
    f = lambda s: dtExact(t) - div( K(exact(s))*grad(exact(s)) )
    g = lambda s: K(exact(s))*grad(exact(s))
```

Task 6 (2 pts)

Solve (7) for $T = 0.25$. This requires a time loop and the choice of an appropriate time step size Δt. Start with something small, e.g. $\Delta t = T/100$ or even $\Delta t = T/1000$.
Note: u^n enters the weak form as a parameters which needs to be updated.

```
while time <= endTime:
    u_h_n.assign(u_h)
    scheme.solve(target=u_h)
    time += scheme.model.dt   # same name on model as the above constant dt
    scheme.model.t = time     # same name on model as the above constant t
```

Task 7 (1 pt)

Implement a Crank-Nicolson time scheme for (7).

A vector valued problem

Consider the Cahn-Hilliard equation with $\psi\colon \mathbb{R} \to \mathbb{R}$ be defined as

$$\psi(x) = \frac{(1-x^2)^2}{4} \text{ and } \phi(x) := \psi(x)' = x^3 - x.$$

The strong form for the solution $u\colon \Omega \times [0,T] \to \mathbb{R}$ is given by

$$\partial_t u - \Delta(\phi(u) - \varepsilon^2 \Delta u) = 0 \quad \text{in } \Omega \times [0,T], \tag{8}$$

$$u(\cdot, 0) = u_0(\cdot) \quad \text{in } \Omega, \tag{9}$$

$$\partial_n u = \partial_n \Big(\phi(u) - \varepsilon^2 \Delta u\Big) = 0 \quad \text{on } \partial\Omega \times [0,T]. \tag{10}$$

The above equation contains 4th order derivatives which cannot be directly discretized with standard Finite Element methods, however, we can introduce an auxiliary variable

$$\eta = \phi(u) - \varepsilon^2 \Delta u$$

to transform the equation into a system of 2 equations only containing derivatives up to 2nd order. We obtain:

$$\partial_t u - \Delta \eta = 0 \quad \text{in } \Omega \times [0,T] \tag{11}$$

$$\eta - \phi(u) + \varepsilon^2 \Delta u = 0 \quad \text{in } \Omega \times [0,T] \tag{12}$$

$$u(\cdot, 0) = u_0(\cdot) \quad \text{in } \Omega, \tag{13}$$

$$\eta(\cdot, 0) = \eta_0(\cdot) = \phi(u_0(\cdot)) - \varepsilon^2 \Delta u_0(\cdot) \quad \text{in } \Omega, \tag{14}$$

$$\partial_n u = 0 \quad \text{on } \partial\Omega \times [0,T], \tag{15}$$

$$\partial_n \eta = 0 \quad \text{on } \partial\Omega \times [0,T]. \tag{16}$$

with Neumann-Null boundary conditions (also called **do-nothing** boundary conditions in this case).
In order to understand the coupling we write this as a vector valued problem, with $\mathbf{u} := (u_0, u_1)^T$ (with $u_0 = u$ and $u_1 = \eta$ in the above equations). For the test functions from our discrete space we have $\mathbf{v} := (v_0, v_1)^T$.
We derive a weak from for this equation in the same way as before by multiplying with a test function, integrating over the domain Ω and applying integration by parts to the terms with 2nd order derivatives. For the time derivative we use an implicit Euler discretization.

We obtain:

$$\int_\Omega \left(\frac{u_0^{n+1} - u_0^n}{\tau}\right) v_0 + \nabla u_1^{n+1} \cdot \nabla v_0 = 0$$
$$\int_\Omega u_1^{n+1} v_1 - \phi(u_0^{n+1}, u_0^n) v_1 - \varepsilon^2 \nabla u_0^{n+1} \cdot \nabla v_1 = 0$$

Note that we also replaced $\phi(u_0^{n+1})$ with $\phi(u_0^{n+1}, u_0^n)$ to achieve energy stability. We follow the approximation by Eyre which reads

$$\phi(v, w) = v^3 - w. \tag{17}$$

We want to solve a vector valued problem and therefore create a Lagrange space over a vector valued range with dimension 2.

```
# space for solution
space = lagrange(gridView, order=1, dimRange=2)

# discrete functions needed for form
u_prev  = space.interpolate([0]*2, name="u_h_prev")
u_h = space.interpolate([0]*2, name="u_h")

u = TrialFunction(space)
v = TestFunction(space)

tau  = Constant(1., name="tau")       # timestep constant
eps  = 0.05
eps2 = Constant(eps**2, name="eps2") # we need eps*eps
```

Task 8 (2 pts)

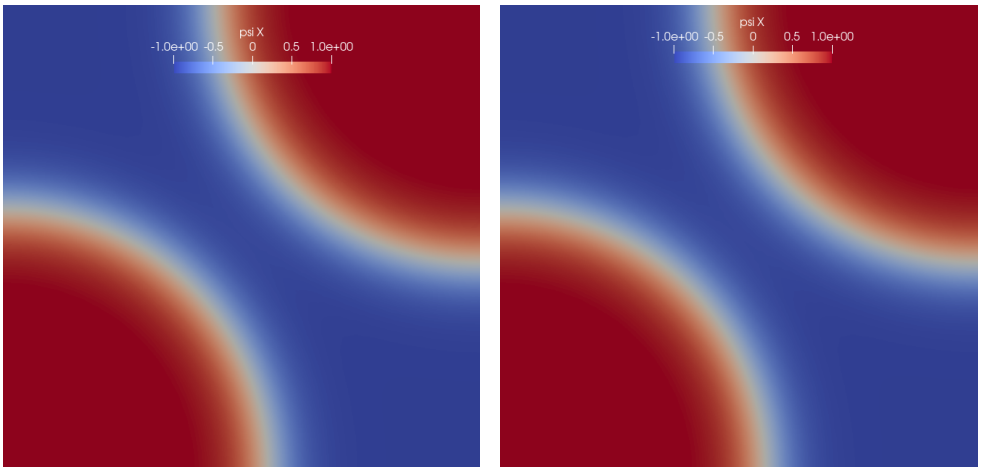
Formulate the above described weak form.
Note: `u[0]` refers to u_0 and `v[0]` will refer to the basis function v_0 associated with the first equation. Accordingly, `u[1]` and `v[1]` will refer to the auxiliary variable u_1 and the basis function v_1 for the second equation.

Task 9 (3 pts)

Solve the equations (11)-(16) with the following initial data

```
from ufl import conditional, sin, pi
def initial(x):
    h = 0.01
    g0 = lambda x,x0,T:
        conditional(x-x0<-T/2,0,conditional(x-x0>T/2,0,sin(2*pi/T*(x-x0))**3))
    G  = lambda x,y,x0,y0,T: g0(x,x0,T)*g0(y,y0,T)
    return 0.5*G(x[0],x[1],0.5,0.5,50*h)
df = space.interpolate(initial, name="solution")
```

until $T = 2$ in $\Omega = [0,1]^2 \subset \mathbb{R}^2$ with a decently fine grid, e.g. more than 5000 elements and $\tau = 1e-3$.



(a) $\phi(v, w) = v^3 - v$ (implicit) (b) $\phi(v, w) = v^3 - w$ (Eyre)

Figure 1: Solution of (11) at $t = T$ for two different choices of ϕ.

Except a final solution similar to Figure 1. This system of equations is very sensitive to perturbations, so your solution might look different, for example, a to large time step might lead to different solutions.

Lycka till!